

Rethinking Feature Quality: Frozen ViT versus HOG under Linear SVM and k-NN Evaluation on STL-10

Yilong Wang Lingxuan Zhu Mingxin Jiang Junyi Li
Technische Universität Berlin

{yilong.wang, lingxuan.zhu, mingxin.jiang, j.li.1}@campus.tu-berlin.de

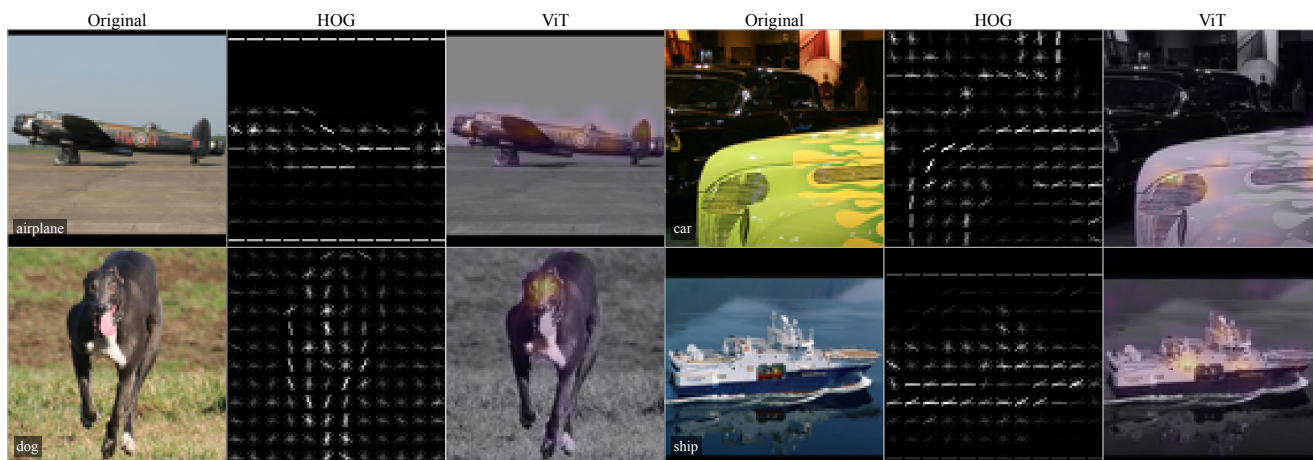


Figure 1. **Qualitative comparison of HOG and ViT feature representations.** For four representative STL-10 images, we show the original crop, the HOG edge descriptor, and the frozen DINOv3 ViT-S/16 attention map overlaid on the image. HOG only captures local edges and gradients, while ViT attention highlights the object itself without any label supervision.

Abstract

We study how much labelled data self-supervised pretraining can save on STL-10, comparing a classical HOG baseline against frozen ViT features from DINOv3 under an identical linear SVM evaluation across varying training-data fractions. The pretrained representation is dramatically more label-efficient, matching HOG’s full-data accuracy with only 1% of the labels, with errors concentrated among visually similar categories that HOG confuses far more broadly. A complementary k-NN evaluation confirms this gap is representational, not classifier-driven.¹

1. Introduction

Image classification has long relied on hand-crafted descriptors such as HOG [1], which capture local gradient statistics

but little semantic structure. The ViT [2] instead learns its representation from data, but typically needs large-scale labelled pretraining to become competitive, leaving a gap for tasks with a limited labelling budget. Self-supervised pretraining closes this gap by learning transferable representations from unlabelled images alone: DINOv3 [3] pretrains ViTs via self-distillation, without requiring any labels. Figure 1 compares the resulting HOG and ViT features on example images, showing that the latter already attend to the object itself without label supervision.

We study how much labelled data such a representation can save on STL-10, comparing HOG features against frozen ViT features from DINOv3 (Figure 2) under a linear SVM trained across varying labelled-data fractions. Our results in §4 show that: **(i)** ViT features are dramatically more label-efficient than HOG features, matching its full-data accuracy with only 1% of the labels; **(ii)** HOG features struggle with visually similar objects far more broadly than ViT features; and **(iii)** ViT features remain linearly

¹Code available at <https://github.com/YilongWangBerlin/hog-vs-vit-stl10>.

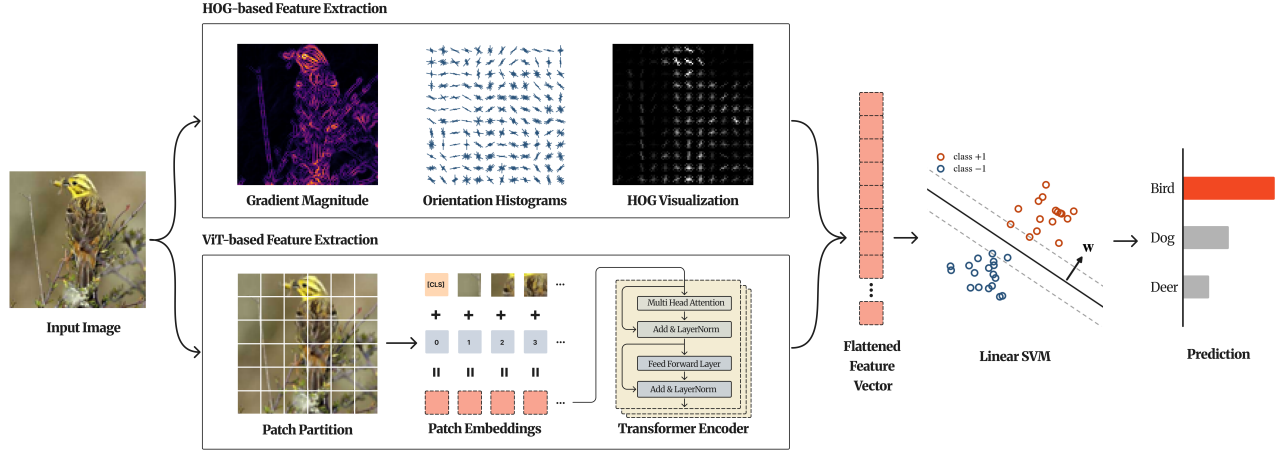


Figure 2. **Comparison framework of HOG and ViT feature extraction.** Given the same input image, we compare two alternative feature extractors, i.e., HOG-based feature extraction and ViT-based feature extraction, while keeping the downstream linear SVM classifier unchanged. The extracted features are converted into a unified feature vector and then fed into the same linear SVM for final prediction.

separable even without a trained classifier, while HOG features degrade sharply under K-NN.

2. Related Work

Visual Representation Learning. Classical pipelines use hand-crafted descriptors such as HOG [1], which encode local gradient statistics but capture little semantic structure. The ViT [2] instead learns its representation from data, splitting an image into patches processed by a transformer encoder. DINOv3 [3] addresses this via self-distillation on unlabelled images, yielding transferable features and providing the frozen encoder used in this work.

3. Technical Information

3.1. Dataset

All experiments are conducted on **STL-10**, a ten-category dataset designed for evaluating visual representation learning under limited supervision. Classifiers are trained on different fractions of the labelled training set (1%, 5%, 10%, 25%, 50%, 100%) and evaluated on the same test set.

3.2. Feature Extractor

As a classical baseline, each image is represented using **HOG features** [1]. HOG descriptors encode local gradient orientation statistics and are directly used to train a downstream classifier in §3.3 without additional preprocessing. Unlike self-supervised representations, whose parameters are learned from data during pretraining, the HOG feature extraction process is entirely handcrafted and involves no learning at any stage. For the self-supervised representation, image features are extracted using a pretrained

DINOv3 ViT-S/16 encoder [3]². The encoder remains frozen throughout the experiments, and no fine-tuning is performed on STL-10. Instead, the extracted feature vectors are ℓ_2 -normalized before being used to train the same linear SVM classifier.

3.3. Downstream Classifier

Both feature representations are evaluated using the same downstream classifier and training settings, so that performance differences reflect the quality of the underlying representations rather than the classifier.

LINEAR SVM. For each labelled-data fraction, a class-balanced subset is drawn via stratified sampling with a fixed random seed, and the classifier is trained with a maximum of 5,000 iterations. The regularization parameter C is selected from $\{0.01, 0.1, 1, 10, 100\}$ via 3-fold cross-validation on the training subset, then evaluated once on the test set.

K-NN. As a parameter-free complement, we apply a k -nearest-neighbour classifier directly on the ℓ_2 -normalized frozen features, retrieving the nearest training examples by cosine similarity, with $k \in \{1, 5, 10, 20, 50, 100\}$, and predicting via similarity-weighted vote over their labels.

4. Results

ViT features outperform HOG features across all label fractions, and are far more label-efficient. Figure 3 summarizes the accuracy of a linear SVM trained on HOG features versus ViT features across labelled-data fractions. ViT features nearly triple HOG features’s accuracy at the

²<https://huggingface.co/facebook/dinov3-vits16-pretrain-lvd1689m>

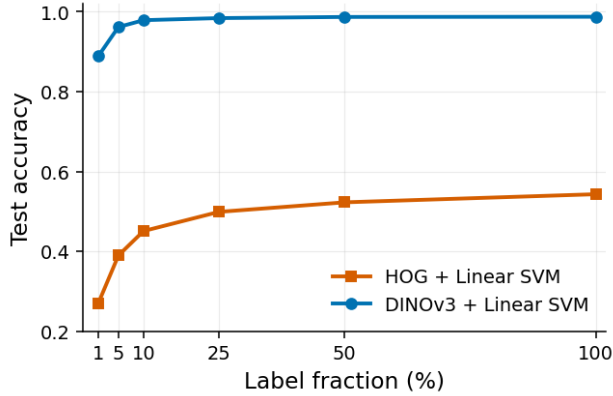


Figure 3. Linear evaluation accuracy on STL-10 using different fractions of labelled training data.

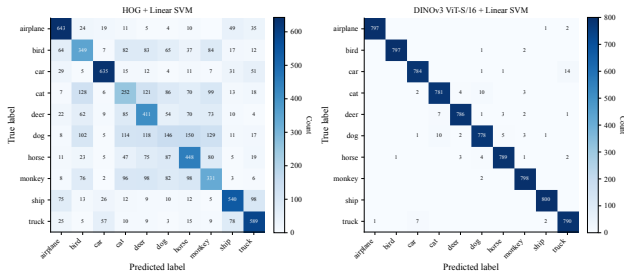


Figure 4. Confusion matrices of the best-performing HOG and ViT classifiers trained on the full labelled training set.

smallest fraction and saturate after just 10% of the labels, while HOG features keep improving without closing the gap. This reflects the different roles of supervision: HOG features depend on labels to fit a boundary, while ViT features are already informative beforehand.

HOG features struggle with visually similar objects, while ViT features do not. Figure 4 shows the confusion matrices of the best-performing linear SVM on ViT features and HOG features. HOG features confuse extensively among the animal categories, while ViT features errors are extremely rare. Gradient descriptors capture object outlines but not the pose and appearance variation that distinguishes similar categories, exactly where ViT features excel.

ViT features remain linearly separable even without a trained classifier. Table 1 compares linear SVM against a parameter-free κ -NN classifier on the same frozen features. ViT features lose almost nothing under κ -NN, while HOG features drop sharply, confirming that the advantage of ViT features lies in the representation itself, not the classifier.

5. Discussion

What linear evaluation measures. Since the encoder is frozen and only a linear boundary is fit, accuracy reflects

Feature extractor	linear SVM	κ -NN (best k)
HOG	54.3%	38.8% ($k=5$)
DINOv3 ViT-S/16	98.8%	98.5% ($k=10$)

Table 1. Test accuracy on STL-10 with the full labelled training set, comparing a trained linear SVM classifier against κ -NN classifier on the same frozen features.

how linearly separable the feature space already is, not what a more expressive classifier could add. The gap in §4 is thus attributable to the representations themselves: ViT features are nearly linearly separable prior to any label supervision, while HOG features encode little class-relevant structure. The κ -NN results corroborate this: ViT features achieve comparable accuracy under linear SVM and κ -NN, whereas HOG features degrade substantially, indicating their nearest neighbours are often of the wrong class.

6. Conclusion

This work compared a classical HOG baseline with frozen DINOv3 ViT-S/16 features under an identical linear SVM evaluation on STL-10. ViT features consistently outperform HOG features, matching its full-data accuracy with only 1% of the labels, and remain robust under κ -NN while HOG features degrade sharply. This advantage is concentrated exactly where hand-crafted gradient features are weakest, namely in distinguishing visually similar categories, indicating that the gap lies in representation quality rather than classifier capacity.

AI Usage Disclosure

We used AI assistance for drafting the report, debugging code, and formatting. All experimental design and interpretation were done by the authors.

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